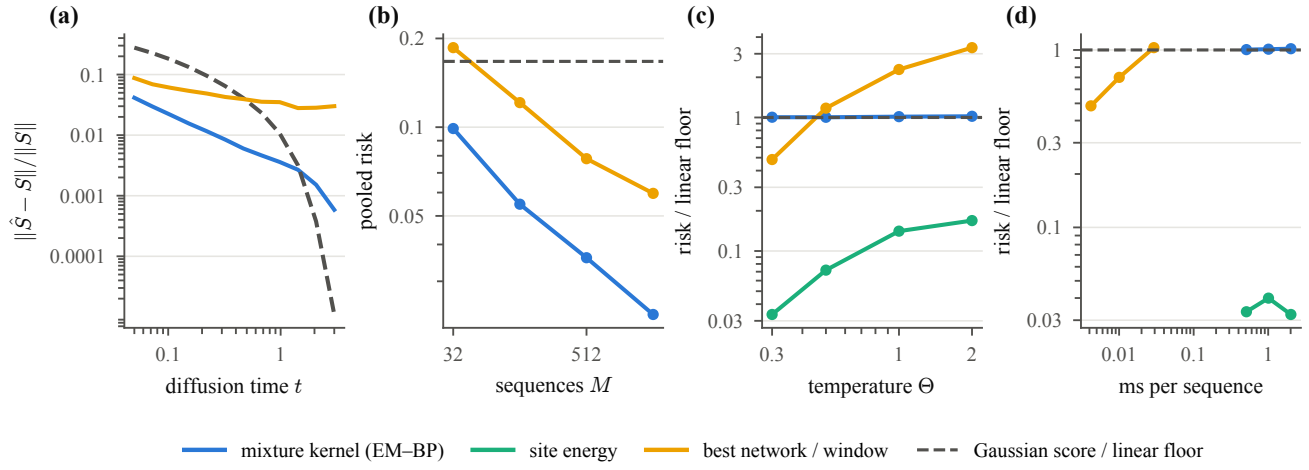
Figures first

The decisive figures of the consolidated note's plan, from frozen outputs

6 October 2026 · branch `claude/phi4-field-2026-10-01` · notation `docs/NOTATION.md`, metrics `docs/METRICS.md`

The consolidated note (section 28) and the call of 25 September ask for five or six decisive figures before the prose, and a summary figure that "really shows everything you want to show". This document is that selection, for choosing: the seven figures of the note's plan, each built from frozen outputs only and named by the experiment that produced it, and the summary. Nothing here is new data except E39 (figure 2); E40 (the FFT transition) and the sixteen-seed replications E35x/E36x are reported in the field report.

In one paragraph. The estimator learns the local law of a Markov chain from noisy sequences by EM and turns it into a score by exact inference on the chain (figure 1); on the Gaussian chain it reproduces the closed-form score and the Kalman smoother to 2×10^{-13} (figure 2). On the Laplace chain the learned mixture kernel beats the Gaussian score wherever non-Gaussian structure survives the noise, and every network at every level (figure 3); with 128 sequences it is more accurate than the best network with 2048 (figure 4). On the ϕ^4 field the same mixture kernel falls onto the linear floor, because its drift is linear, while a five-parameter site energy reaches 3–17 % of the floor at $M = 2048$ and the networks are worse than the floor from $\Theta = 0.5$ on (figure 5 and the field report). Grid BP costs 0.5–2 ms per sequence, 6 to 120 times a network's, for an error 14 to 32 times smaller (figure 6), and an FFT transition does not change that at these grid sizes (E40). Under a global latent the first-order kernel is 1.5–1.7 times worse than the linear floor, and the same kernel on the augmented state repairs it (figure 7).



Summary figure. (a) Per-level score error on the Laplace chain at $M = 2048$ (E28). (b) Pooled risk against the number of training sequences on the Laplace chain (E02a, E03). (c) Risk over the linear floor across temperature on the ϕ^4 field at $M = 2048$ (E33). (d) Risk over the linear floor against inference cost on the field at $h = 1$, 0.5, 0.25 (E33, E24x). Blue: the paper's mixture kernel (EM–BP); green: the site energy; yellow: the best network in (a, b), the window head in (c, d); dashed: the Gaussian score or linear floor.

1. The method

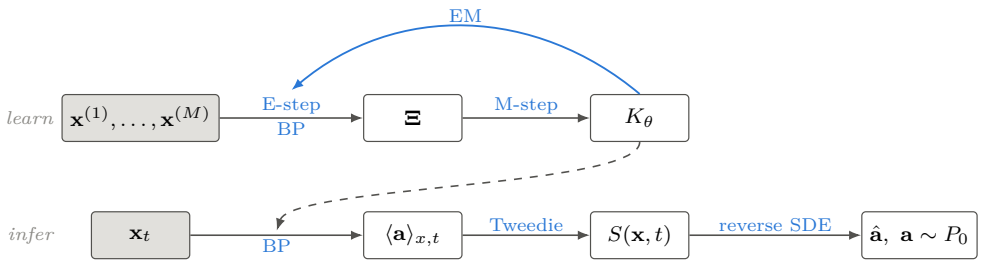


Figure 1: **Learn the local law, infer globally.** Top: EM on M noisy sequences; the E-step runs BP and sums its pair marginals into Ξ, the M-step updates the kernel, and the two alternate. Bottom: the learned kernel inside BP at any level t gives the posterior mean, Tweedie's identity turns it into the score, and the reverse SDE uses the score to denoise or generate. A direct score network learns all of the bottom row at once; here only the kernel is learned.

2. The Gaussian sanity check

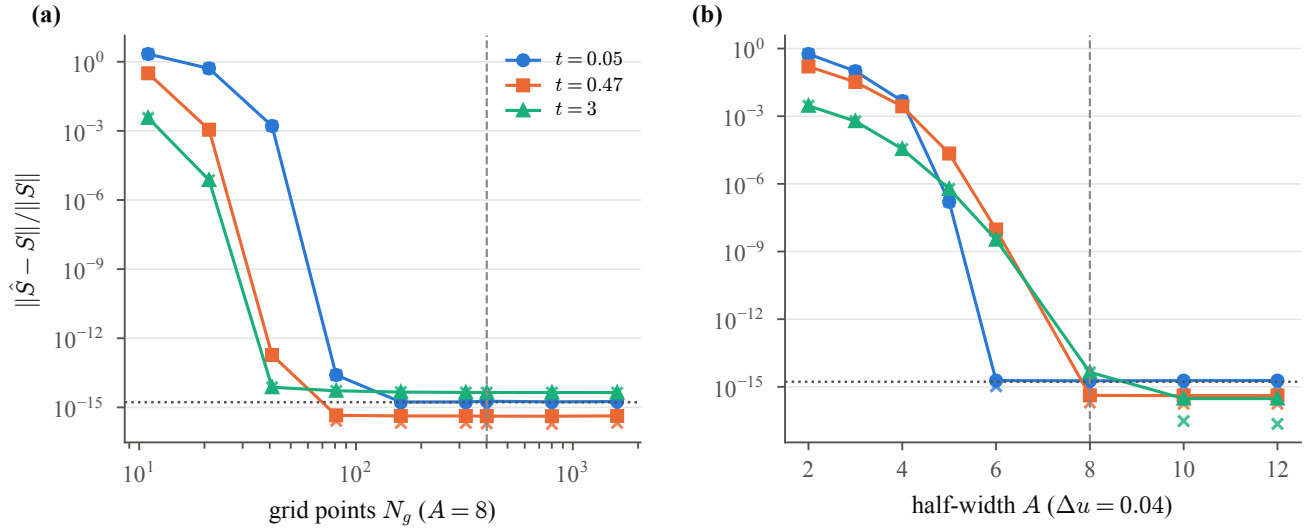


Figure 2: **Grid BP converges to the closed-form score and to the Kalman smoother** (E39, Gaussian linear chain, $\rho = 0.85$, $L = 32$, 256 chains, frozen). Pooled relative score error against the closed form (lines) and against an independent Rauch–Tung–Striebel smoother (crosses), at three of the twelve levels. (a) Refining the grid at $A = 8$; (b) widening the domain at the operating spacing. Dashed: the operating grid; dotted: the smoother against the closed form.

The smoother agrees with the closed form to 1.7×10^{-15}, and grid BP at $(401, 8)$ with both to 1.9×10^{-13} at every level. Refining the grid converges faster than any power of the spacing and reaches 10^{-10} by $N_g = 81$ at the smallest levels and 41 at the largest; at $A = 3$ the truncation costs 9.9×10^{-2}. The paper now states this as a proposition with its proof (appendix B). One registered clause failed as written: moment-projected BP on the same grid is off by up to 0.26, because the grid truncates its backward messages, whose variance grows like e^{2t}; widened to $A = 64$ it is exact to 10^{-15} up to $t = 1.4$ (E39).

3. Where non-Gaussian structure pays

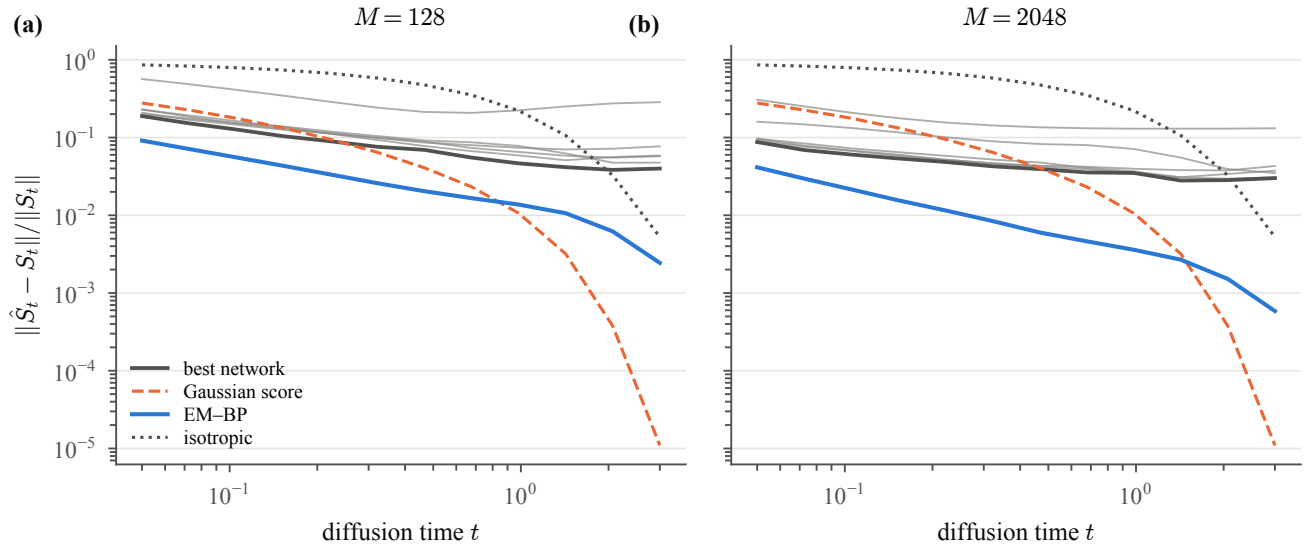


Figure 3: **The gain from non-Gaussian structure, level by level** (E28, the Laplace linear chain, $\rho = 0.85$, $L = 32$, sixteen seeds, frozen). Relative score error against the exact score at each level for EM–BP ($C = 8$), the Gaussian score (the LMMSE of the true covariance), the six networks of the paper (thin) and the best of them at each level, and the isotropic prior. (a) $M = 128$; (b) $M = 2048$ training sequences.

EM–BP is below the best network at every level and both sizes (2 to 47 % of its error at $M = 2048$), and below the Gaussian score until the noise has erased the non-Gaussian structure: the two cross near $t = 1$ with 128 sequences and near $t = 2$ with 2048, the crossover moving with data as E28 found. At $t = 0.05$ and $M = 2048$ the three errors are 0.041 (EM–BP), 0.088 (best network) and 0.278 (Gaussian score).

4. Sample efficiency

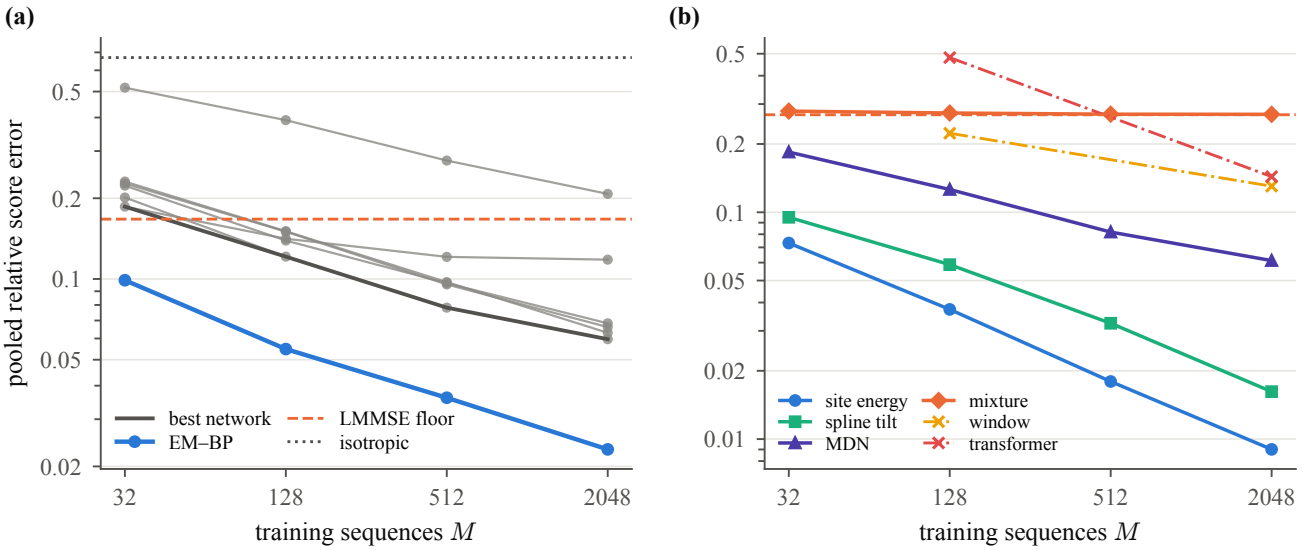


Figure 4: **Error against the number of training sequences.** (a) The Laplace linear chain (E03 networks, E02a EM–BP with $C = 8$; the paper's baselines table), schedule-pooled risk, sixteen seeds; dashed: the LMMSE floor; dotted: isotropic. (b) The ϕ^4 field at $\Theta = 0.3$, $h = 1$ (E33): the four kernels (risk selection) and the two networks; dashed: the linear floor.

On the Laplace chain EM–BP is below the LMMSE floor already at $M = 32$ (0.099 against 0.167), where every network is above it, and with 128 sequences (0.055) it beats the best network with 2048 (bidirectional GRU, 0.060). On the field the site energy falls from 0.073 to 0.009 between $M = 32$ and 2048 and the spline from 0.095 to 0.016, while the mixture kernel stays on the floor (0.27) at every size; the networks, trained at $M = 128$ and 2048, reach 0.13 (window) and 0.14 (transformer).

5. The temperature sweep

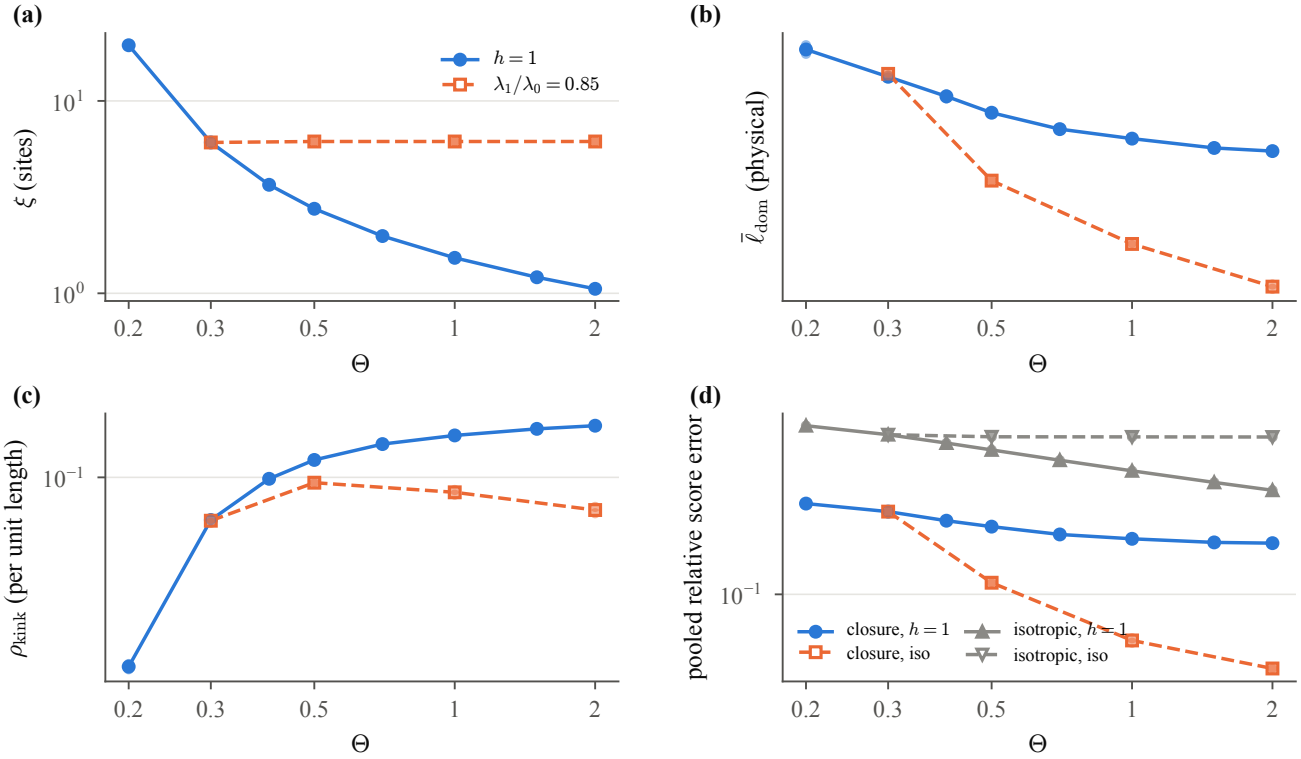


Figure 5: **The ϕ^4 field across temperature** (E32 fixed-spacing and iso-correlation sweeps, E34; the field report's figure). Correlation length, domain length, kink density and the score error of the linear closure across Θ.

At fixed spacing ($h = 1$) the linear floor falls only from 0.30 to 0.18 over $\Theta = 0.2$–2 while kinks multiply (0.016 to 0.166 per bond) and domains shorten (9.0 to 5.3 sites); along the iso-correlation line it falls from 0.269 to 0.041 (E32, frozen). The learned arms across Θ are panel (c) of the summary. In one dimension there is no phase transition at $\Theta > 0$; the low-temperature end is a sharp crossover.

6. Refinement and runtime

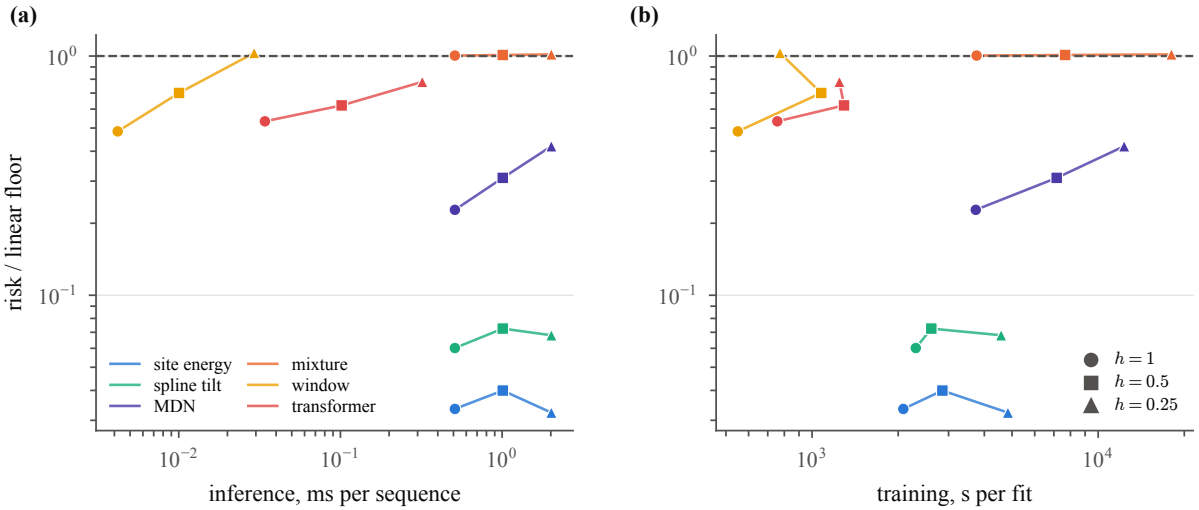


Figure 6: **Accuracy against cost across the lattice spacing** (E33 risk at $M = 2048$ over its linear floor, sixteen seeds; inference from E24's exclusive-node sweep, 8 BLAS threads; training wall clock from E33, shared nodes). Markers: $h = 1$, 0.5, 0.25 ($L = 32$, 64, 128). (a) Inference, ms per sequence; (b) training, s per fit.

Grid BP costs 0.5, 1.0 and 2.0 ms per sequence at $L = 32$, 64, 128; the networks are 6 to 120 times cheaper but sit at 48–103 % of the linear floor, against 3–4 % for the site energy and 6–7 % for the spline. A network trains in 9–22 minutes, an EM fit in 35 minutes to 5 hours (shared nodes). The FFT transition the note proposes (section 9.2; E40) is accurate to second order (6.7×10^{-4} from dense BP at $N_g = 401$) and 5.8 times slower than the dense one there on one core; tuned, it breaks even only near $N_g = 3200$.

7. Misspecification and its repair

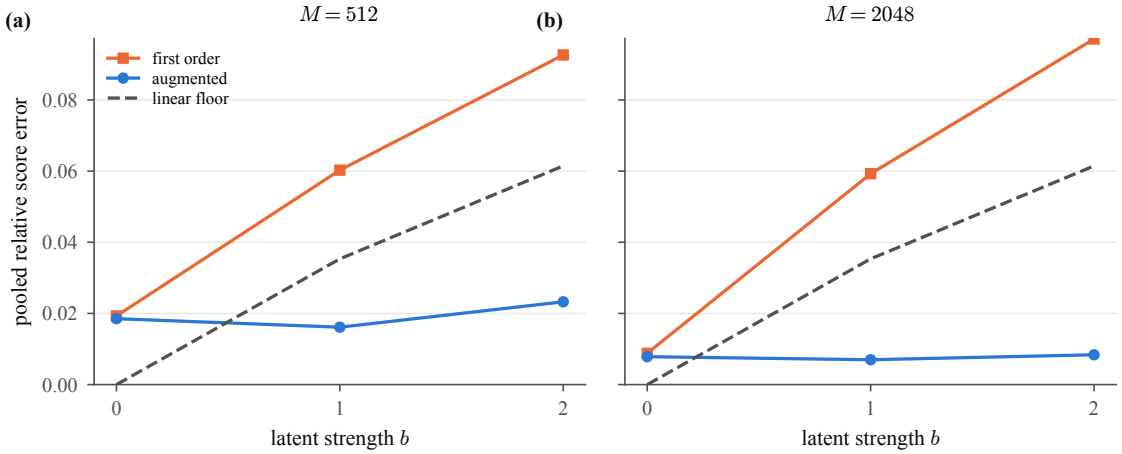


Figure 7: **A global latent the first-order kernel cannot see, and the repair by state augmentation** (E35: $X_i = Gb + U_i$, $G = \pm 1$ per sequence, U the Gaussian linear chain at $\rho = 0.85$; closed-form exact score; two seeds, frozen). The mixture kernel on X (first order) and the same kernel on the state (X_i, G) tied by reflection (augmented), against the linear floor, over the latent strength b. (a) $M = 512$; (b) $M = 2048$.

At $b = 0$ the arms agree and the floor is the exact score (the chain is Gaussian). With the latent the first-order kernel's risk grows with b and does not fall when M quadruples: it is 1.5 to 1.7 times worse than the linear floor at $b = 1$ and 2. The augmented kernel, with the same parameter count, stays flat; at $b = 2$, $M = 2048$ its risk is 8.6 and 17.9 times smaller than the first-order one in the two seeds. Sixteen seeds (E35x) are queued on the cluster, and the figure will be redrawn from them.